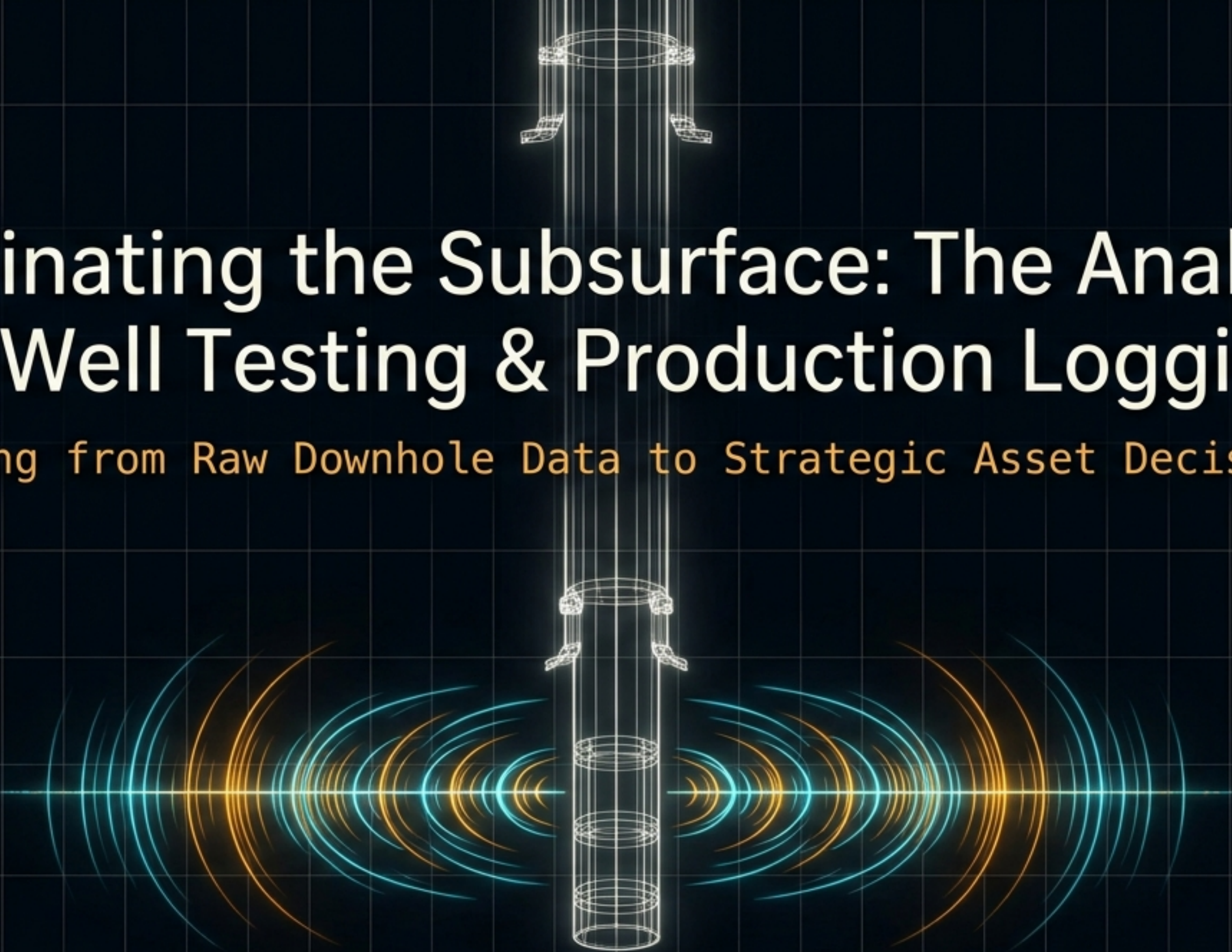


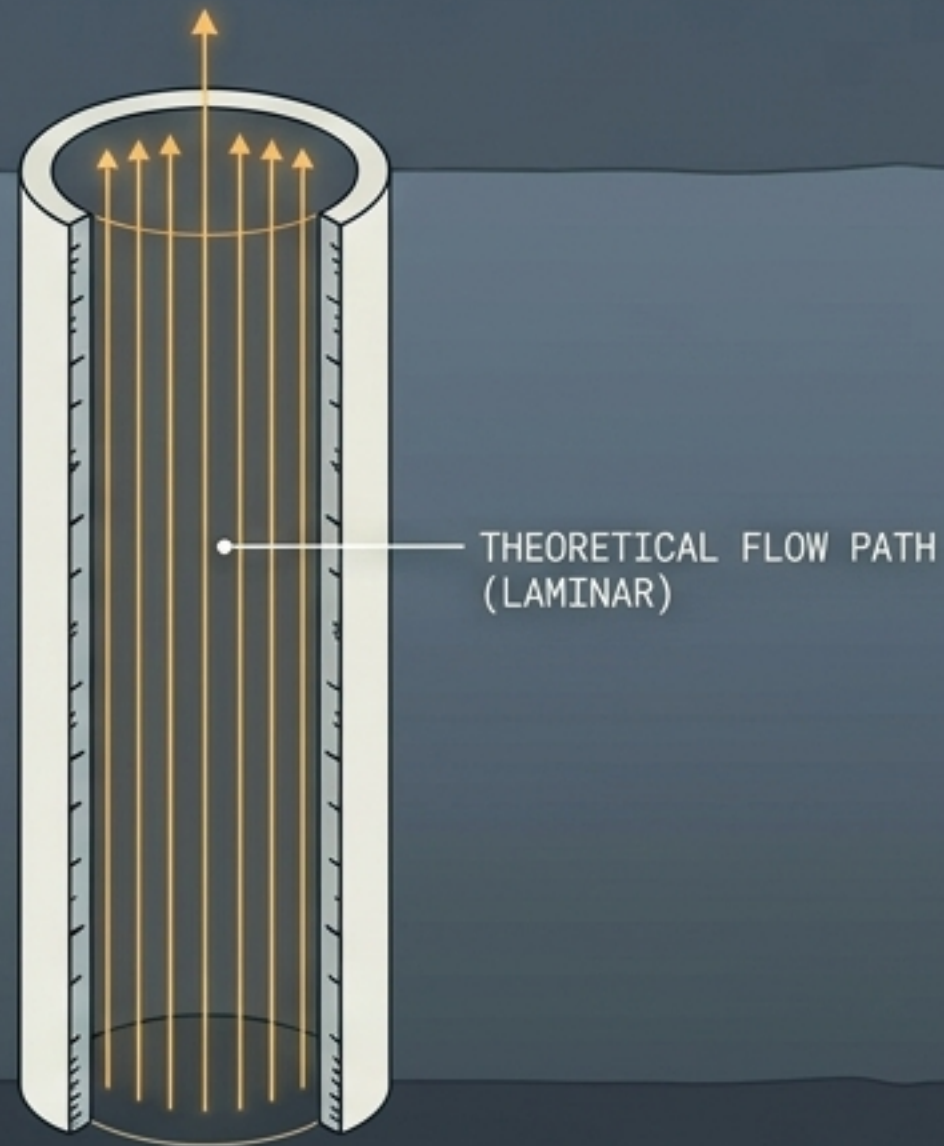


Illuminating the Subsurface: The Analytics of Well Testing & Production Logging

Moving from Raw Downhole Data to Strategic Asset Decisions

The High Cost of Operating a Blind Well

The Assumption



Without data, we manage a theoretical model.

The Reality



Ignorance guarantees sub-optimal recovery.

Moving from Physical Reality to Human Inference



1. Physical Reality

10,000 feet underground,
250°F, complex multiphase
fluid mixing.



2. The Sensor

Hardware interacting with
the hostile subsurface
environment.



3. The Proxy Data

We don't 'see' water or
leaks; we record electrical
capacitance and acoustic
noise frequencies.



4. The Human Inference

The engineer synthesizes
the proxies to calculate
water cut or diagnose
cement channeling.

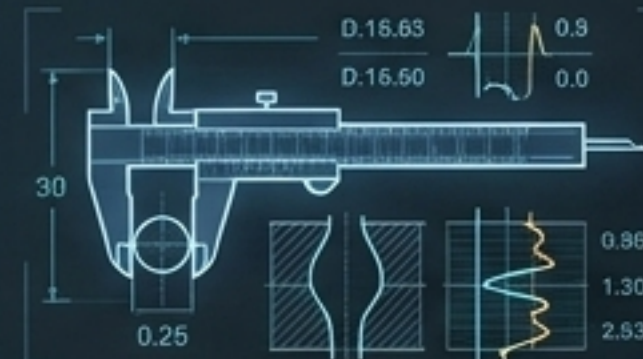
The Four Fundamental Pillars of Subsurface Diagnostics

PRESSURE (P)



The driving force.
Sourced via Gauges, RFT,
and MDT tools.

GEOMETRY (G)



The physical bounds.
Sourced via Caliper and
Gamma Ray tools.

FLOW VELOCITY (V)



The kinetic energy.
Sourced via the Spinner
Flowmeter.

FLUID IDENTITY (F)



The composition.
Sourced via Density (Holdup)
and Capacitance tools.

Classifying the Diagnostic Ecosystem

The Dynamic State (Production Logging Tools)



Spinner Flowmeter



Capacitance /
Water Holdup



Pressure Gauge



Gamma Ray



Temperature
Sensor



Caliper



Fluid Density
(Holdup)



Noise Log

The Static and Transient State (Formation Testing)



RFT (Repeat Formation
Tester)



MDT (Modular Formation
Dynamics Tester)



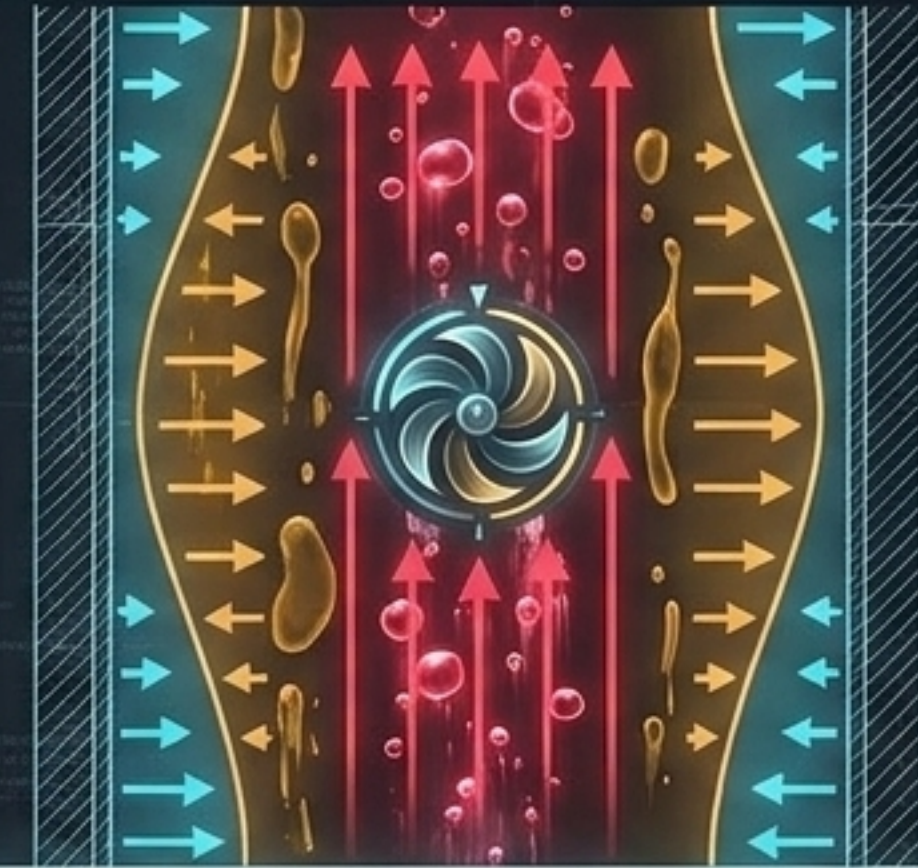
DST (Drill Stem Test)

The Multiphase Reality Behind the Spinner Flowmeter

The Questioned Assumption



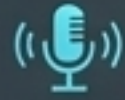
The Multiphase Reality



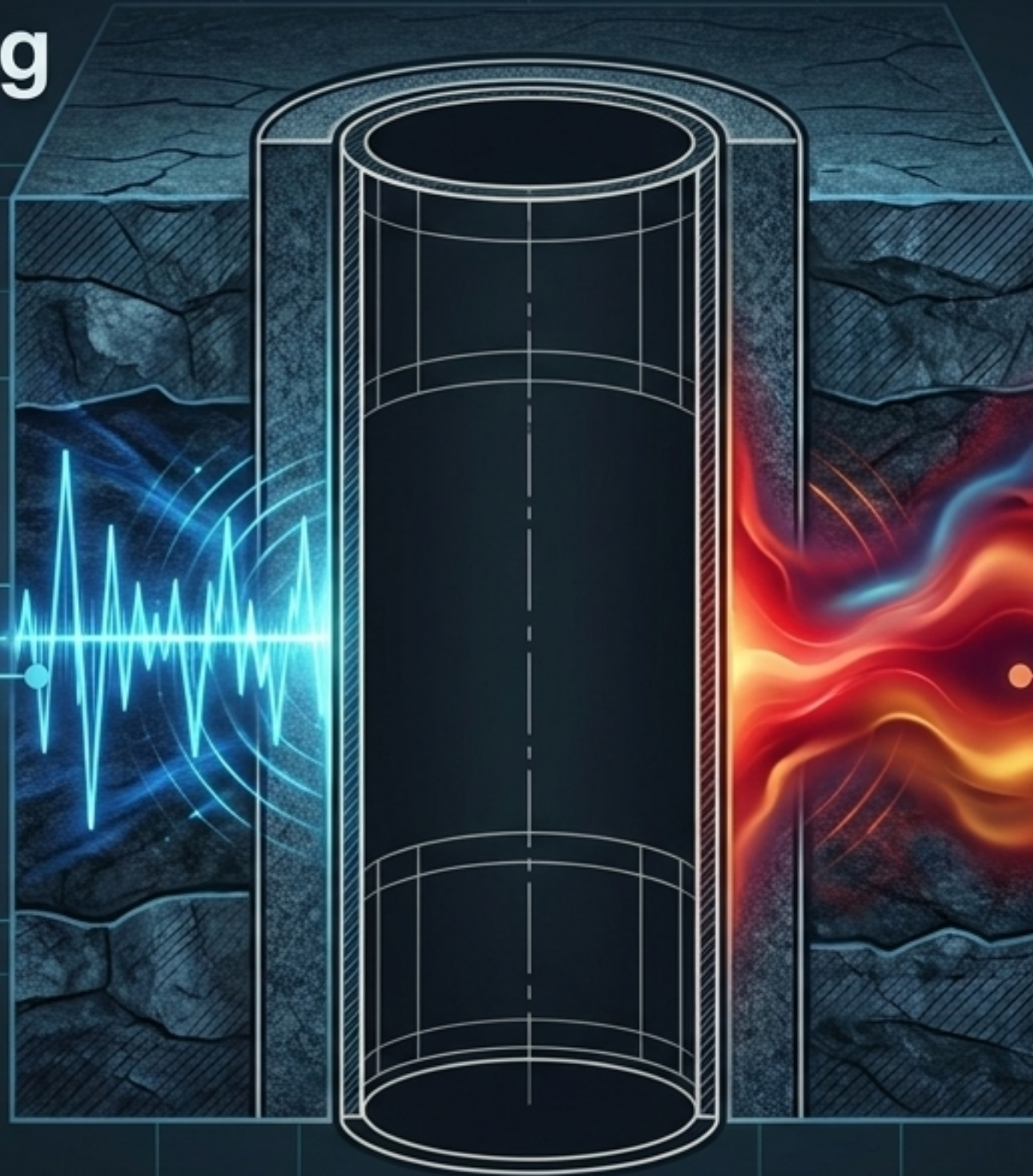
The spinner only measures fluid flow velocity in the immediate path of the impeller. Assuming a uniform velocity profile in multiphase flow leads to drastically flawed production allocations.

Detecting Hidden Anomalies Behind the Casing

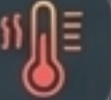
Noise Logs



Identifying distinct acoustic frequencies to detect fluid channeling and movement behind casing, completely invisible to internal flowmeters.

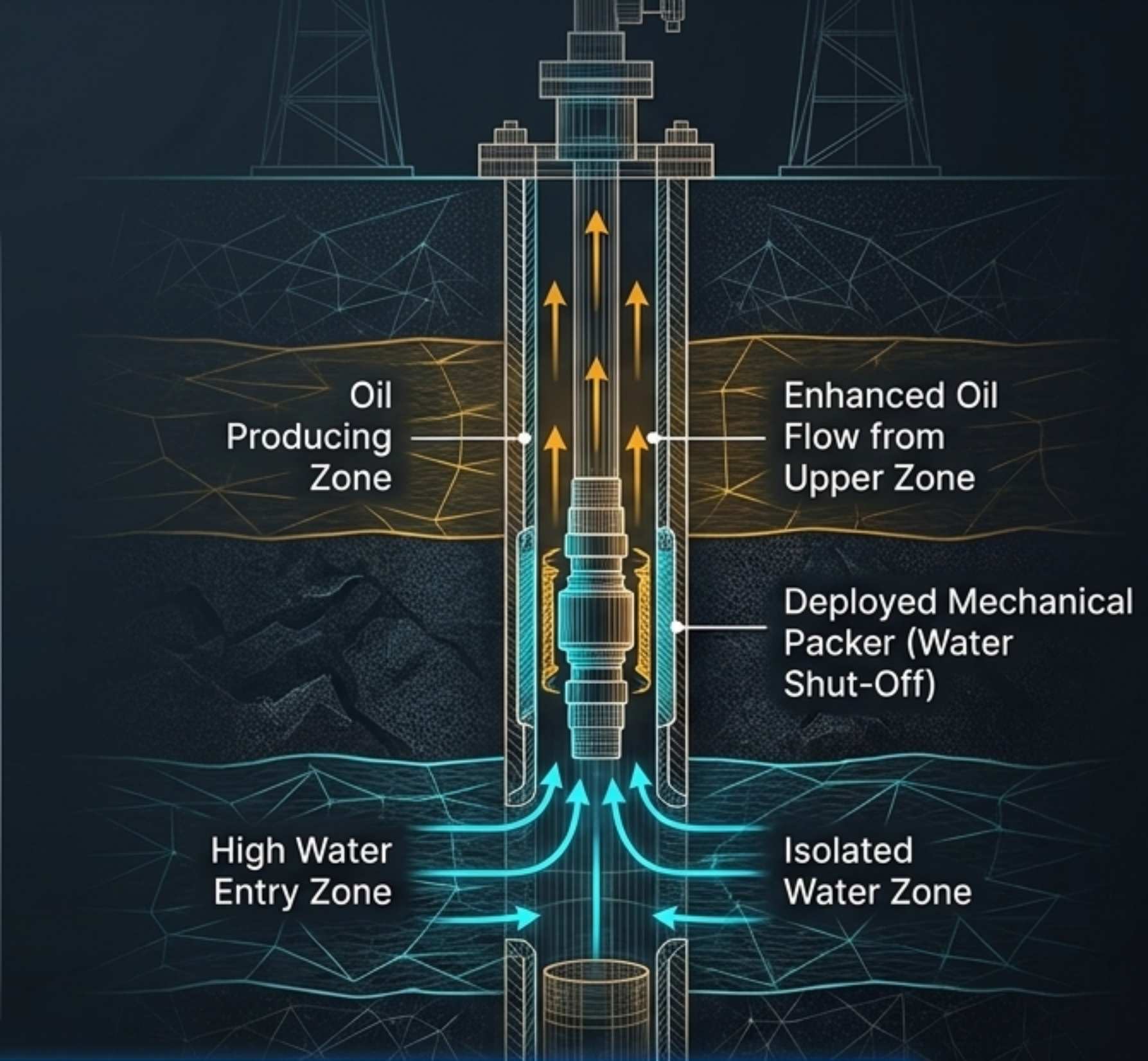
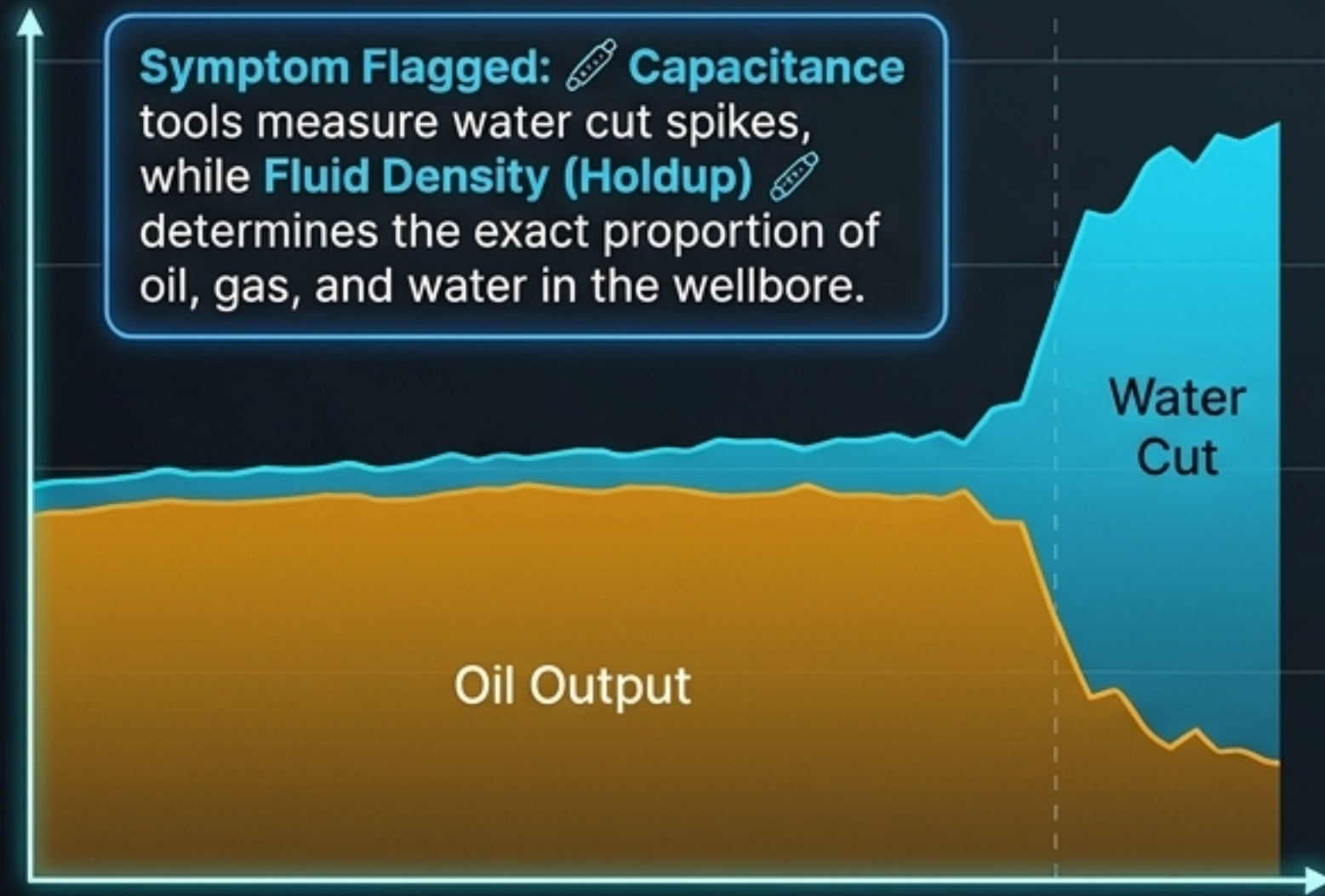


Temperature Sensors



Mapping the Joule-Thomson effect. A sudden temperature drop reveals a gas expansion leak long before a pressure drop registers at the surface.

Reframing Water Cut as an Optimization Trigger



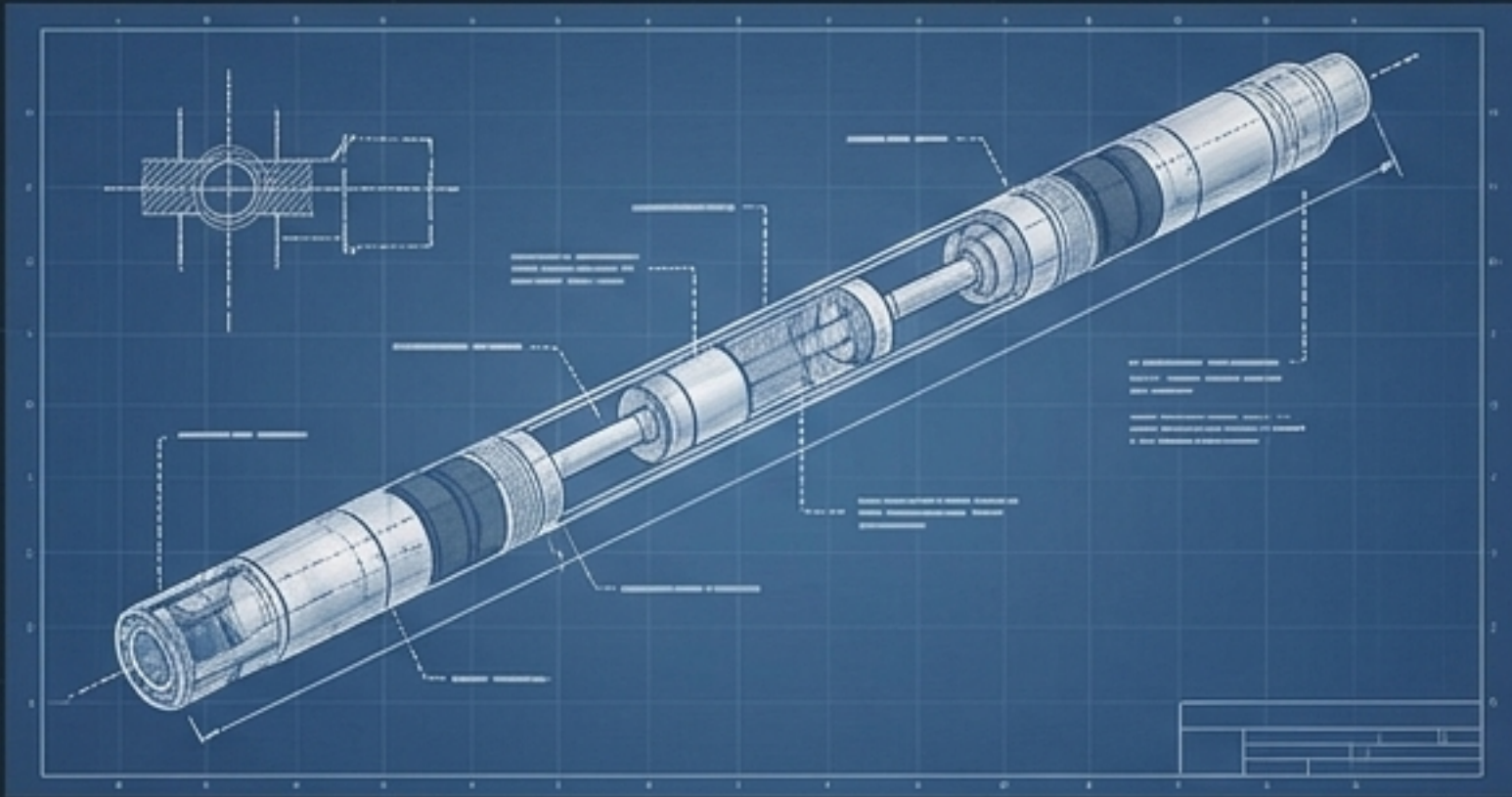
Precision holdup data differentiates between a rising, unmanageable aquifer and a localized, fixable casing leak—turning a 'dead well' into a prime optimization candidate.

Advancing Formation Testing from RFT to MDT



RFT (Repeat Formation Tester)

The Baseline

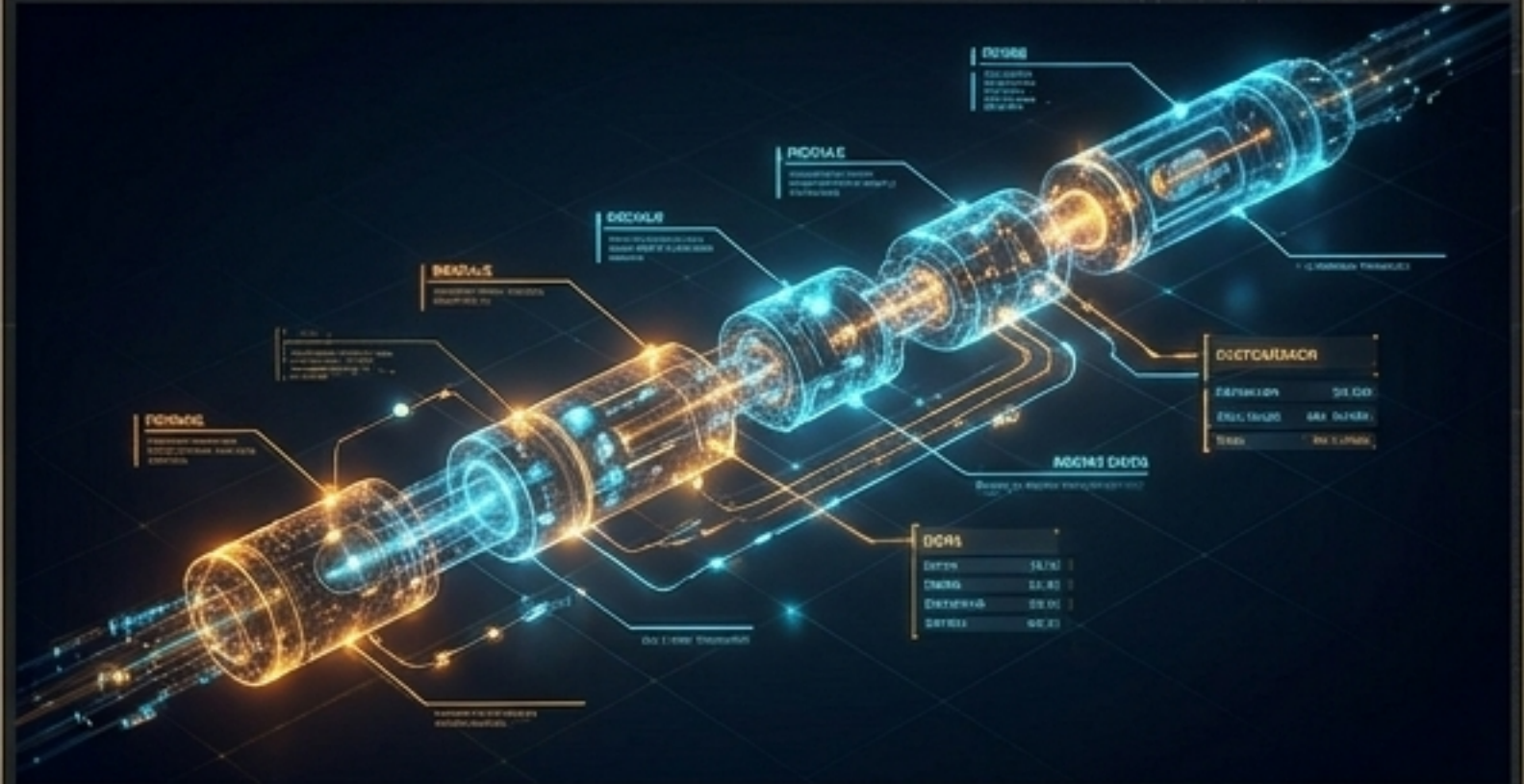


Measures formation pressure and collects fluid samples from individual zones without significant formation damage.



MDT (Modular Formation Dynamics Tester)

The Advanced Evolution



Advanced formation pressure testing and fluid sampling with significantly higher accuracy and modular customization for complex transients.

Proving the Macroscopic Reality with Drill Stem Tests

ultra-thin precision grid

**DST (Drill Stem Test)
Investigation Radius**

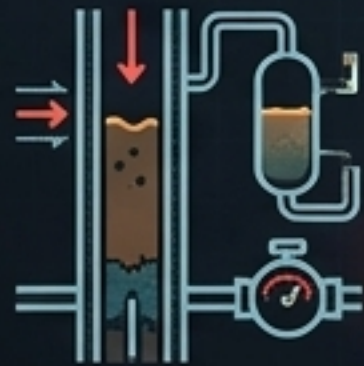
**RFT / MDT
Investigation Radius**

DST evaluates reservoir productivity, pressure, permeability and fluid characteristics before completion. It separates wireline micro-theory from dynamic flowing reality.

280,000	200,000	100	1,000	2,000
PROBE	FLOWER	PODCHGO	TORNO	100L
		DAMK		

• Average Norm 5220 105 51020 1000-511-75000

Three Distinct Perspectives on a Single Pressure Drop



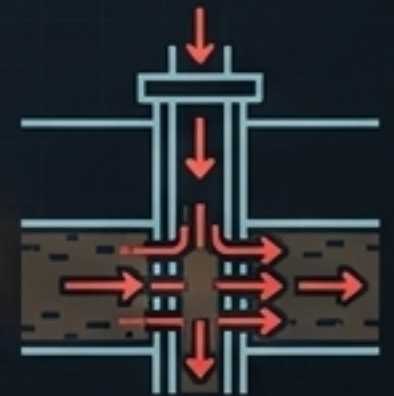
THE PRODUCTION ENGINEER

"Look at the fluid density data. We have heavy emulsion near the wellbore. The formation is fine; we just need to stimulate and clean it."



THE GEOLOGIST

"We've hit a fault block. The reservoir is compartmentalized and physically smaller than we mapped."



THE RESERVOIR ENGINEER

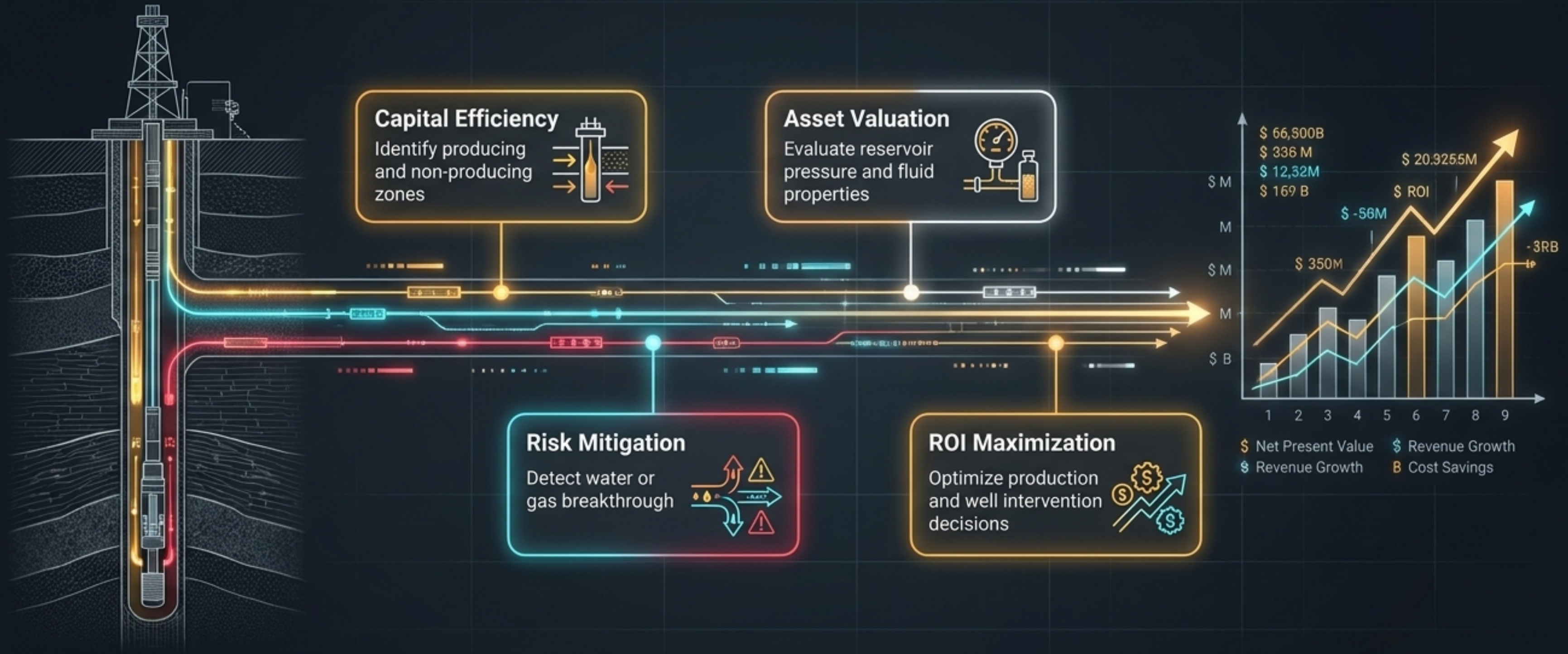
"No, the permeability is just lower than expected. The skin factor is choking the flow near the wellbore."

| The Diagnostic Synergy of the Tool String



No single tool provides absolute truth. Accurate diagnosis relies entirely on the intersection of multiple, independent measurements cross-verifying one another.

Driving the Financial and Strategic Needle



Transitioning to the Digital Subsurface



From Episodic to Continuous

Moving past running a mechanical wireline tool once a year to streaming real-time downhole data.



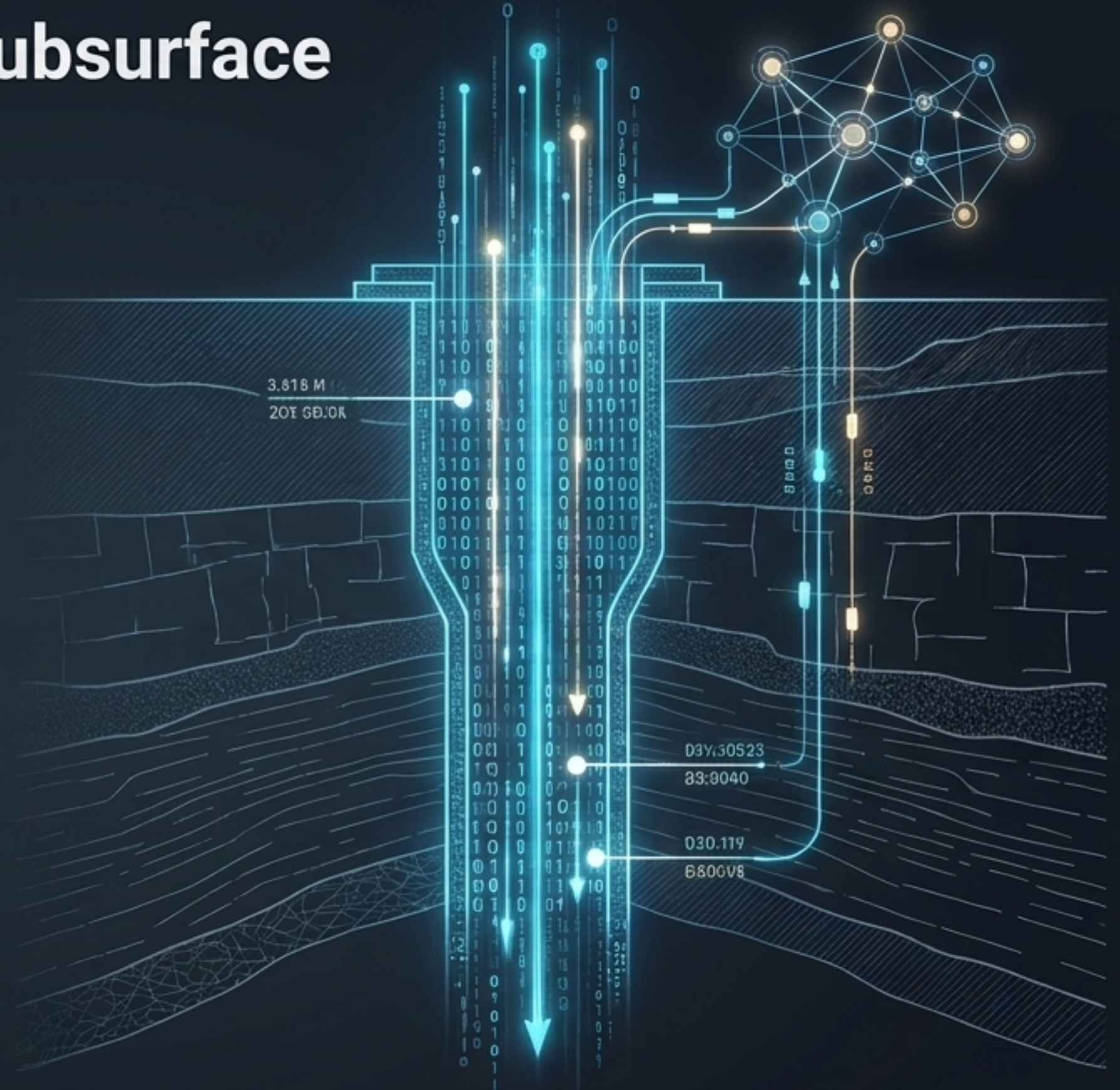
Permanent Downhole Gauges (PDGs)

Deploying permanent architecture to capture infinite, uninterrupted pressure and thermal data streams.



Predictive Analytics

Using AI to replace reactive diagnostics with forward-looking anomaly detection based on continuous fiber-optic acoustic and thermal sensing.



Measure. Analyze. Optimize. Maximize.

The right downhole data leads to the **right production decisions**. Every successful well starts with accurate measurements, **rigorous questioning of assumptions**, and multi-disciplinary synthesis.

DATA DRIVEN DECISIONS. BETTER WELLS. HIGHER RECOVERY. LOWER COSTS.